

STUDENT'S NAME: _____
TEACHER'S NAME: _____

BAULKHAM HILLS HIGH SCHOOL

YEAR 11

YEARLY EXAMINATION

2007

MATHEMATICS
EXTENSION 1

*Time allowed - Two hours
(Plus five minutes reading time)*

DIRECTIONS TO CANDIDATES:

- Attempt **ALL** questions.
- Start each of the questions on a new page.
- All necessary working should be shown.
- Write your teacher's name and your name on the cover sheet provided.

QUESTION 1 (12 marks)

Marks

- (a) Solve the inequality $\frac{3x - 2}{x} \geq 1$ 3
- (b) Find the value of k if $(x - 2)$ is a factor of
 $P(x) = x^4 - 3x^3 + kx^2 - 4$ 2
- (c) If P is the point $(-3, 5)$ and Q is the point $(1, -2)$ find the coordinates of the point R which divides PQ externally in the ratio 3 : 2. 2
- (d) Find the obtuse angle between the lines $2x - y = 0$ and $x - 2y = 0$ giving your answer correct to the nearest degree. 3
- (e) Differentiate $(\sqrt{x} + 3)\left(\frac{1}{\sqrt{x}} - 2\sqrt{x}\right)$ 2

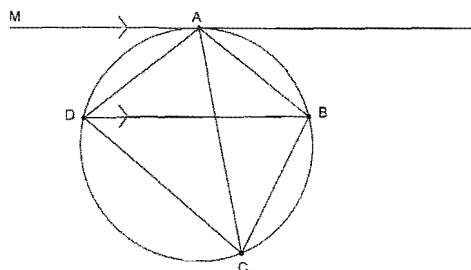
QUESTION 2 (12 marks)

- (a) If $\sin 2A = \frac{1}{2}$ find the exact value of $\frac{1}{\sin A \cos A}$ 2
- (b) The equation $x^3 - 6x^2 + 4x + 2 = 0$ has three real roots α , β and γ .
(i) Write down the values of $\alpha + \beta + \gamma$, $\alpha\beta + \beta\gamma + \alpha\gamma$ and $\alpha\beta\gamma$ 1
(ii) Hence find the value of $(\alpha - 2)(\beta - 2)(\gamma - 2)$ 3
- (c) Prove using mathematical induction that $\sum_{r=1}^n \frac{1}{r(r+1)} = \frac{n}{n+1}$ 4
- (d) Simplify $\frac{xy^{-1} - yx^{-1}}{x - y}$ 2

QUESTION 3 (12 marks)

Marks

(a)



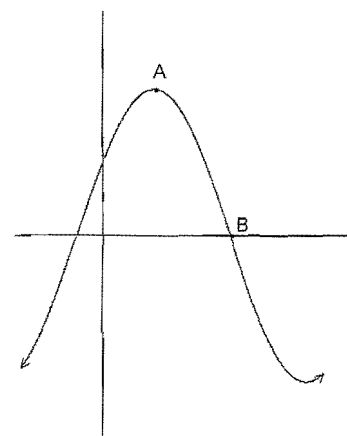
ABCD is a cyclic quadrilateral. The tangent to the circle at A is parallel to DB.

- (i) Copy the diagram 1
- (ii) Give a reason why $\angle ACD = \angle MAD$ 1
- (iii) Give a reason why $\angle ACB = \angle ADB$ 1
- (iv) Hence show that AC bisects $\angle BCD$ 2
- (b) (i) Show that $1 - \frac{2}{x-2} = \frac{x-4}{x-2}$ 1
- (ii) Show that the function $f(x) = 1 - \frac{2}{x-2}$ is increasing for all values of x in its domain. 2
- (iii) Sketch the graph showing clearly the coordinates of any points of intersection with the x and y axis and the equations of any asymptotes. 2
- (c) Use the definition $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ to find the derivative of $f(x) = \frac{1}{x}$. 3

QUESTION 4 (12 marks)

Marks

- (a) (i) Express $\sqrt{3} \sin x + \cos x$ in the form $R \sin(x + \alpha)$ where $R > 0$ and $0 < \alpha < 90^\circ$. 2
- (ii) The graph of $y = \sqrt{3} \sin x + \cos x$ is shown below. Hence, or otherwise, find the coordinates of A and B if A is a maximum turning point and B is where the curve cuts the x axis. 3

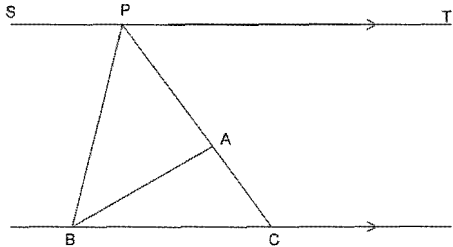


- (b) (i) Show that $\tan(45^\circ + A) = \frac{\cos A + \sin A}{\cos A - \sin A}$ 2
- (ii) Hence show that $\tan(45^\circ + A) = \frac{1 + \sin 2A}{\cos 2A}$ 2
- (c) Solve the equation $\cos^2 x + \sin 2x = 0$ for $0 \leq x \leq 360^\circ$, giving your answer to the nearest degree. 3

QUESTION 5 (12 marks)

Marks

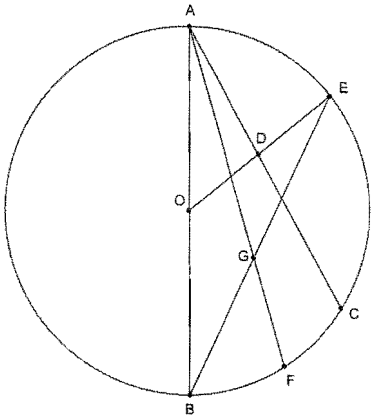
(a) In the diagram, $ST \parallel BC$ and $\angle SPB = \angle PAB$.



- (i) Prove that $\triangle PBA$ is similar to $\triangle PCB$ 2
- (ii) Deduce that $PB^2 = PA \times PC$ 2
- (b) (i) Sketch the polynomial $p(x) = (x + 7)(1 - x)(x - 3)^2$ 1
- (ii) Hence, for what values of x is $p(x) > 0$ 1
- (c) If the equation $3x^3 - 4x^2 + 2x + 1 = 0$ has roots α, β and γ , find the equation with roots $2\alpha, 2\beta$ and 2γ 3
- (d) (i) Show that the sum:
$$a - ap + ap^2 - ap^3 + ap^4 + \dots + ap^{2k} = a \frac{(1 + p^{2k+1})}{1 + p}$$
 2
- (ii) Hence, or otherwise, find an expression for the sum:
$$3^n - 3^{n+1} + 3^{n+2} - 3^{n+3} + \dots + 3^{3n}$$
 1

QUESTION 6 (12 marks)

- (a) Factorise completely: $x^2 + y^2 + 2xy - z^2$. 1
- (b)

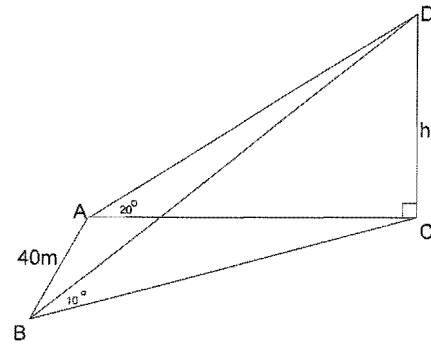


- In the figure above, AOB is the diameter of a circle centre O . D is a point on the chord AC such that $DA = DO$ and OD is produced to E . AF is the bisector of $\angle BAC$ and cuts BE in G .
- (i) Prove that $GA = GB$. 3
- (ii) Prove that $AOGE$ is a cyclic quadrilateral. 2
- (iii) If CE is joined, prove that $CE \parallel AF$. 2
- (c) Use mathematical induction to show that $5^n + 2(11^n)$ is a multiple of 3 for all positive integers n . 4

QUESTION 7

Marks

- (a) The new Baulkham Hills High School pole, CD, of height h metres, near the office, stands with base C on horizontal ground. A is a point on the ground due west of C and B is a point on the ground 40 metres due south of A. From A and B the angles of elevation to the top of the pole are 20° and 10° respectively.



- (i) Find expressions for AC and BC in terms of h . 1
- (ii) Show that $h = \frac{40}{\sqrt{\cot^2 10^\circ - \cot^2 20^\circ}}$ 2
- (iii) Hence find the height of the pole. 1
- (b) A car leasing company provides finance to customers. Clients can borrow \$P at $r\%$ per month reducible interest calculated monthly. The loan is to be repaid in equal monthly payments of \$M.

Let $R = 1 + \frac{r}{100}$ and let A_n be the amount owing after n monthly repayments have been made.

- (i) Write an expression for the amount owing after two repayments (i.e. A_2) in terms of P , R and M . 1
- (ii) Show that the amount owing after the n^{th} repayment is given by $A_n = PR^n - \frac{M(R^n - 1)}{R - 1}$ 2
- (iii) If the amount owing after the n^{th} repayment is $K\%$ of the amount borrowed, show that: 3
- $$R^n = \frac{PK(R - 1) - 100M}{100[P(R - 1) - M]}$$
- (iv) Hence use your calculator to estimate the number of years required for the amount owing to fall to 20% of the amount borrowed if a client borrows \$40,000 and makes monthly repayments of \$800. Interest is charged at 9% per annum compounding monthly. 2

EXTENSION 1

12/11 SOLUTIONS YEAR 11/12

1(a) $\frac{3x-2}{x} \geq 1$ where $x \neq 0$

$3x-2 \geq x$

$3x-2x \geq x$

$2x-2x \geq 0$

$2x(x-1) \geq 0$

$\therefore x \leq 0$ and $x \geq 1$

①

(b) $P(x) = 0$ by remainder theorem

$0 = 2^2 - 3 \times 8 + 4k - 4$

$4k = 12$

$k = 3$

①

(c) $P(-3, 5)$ $Q(1, -2)$

Point B $\left(\frac{2 \times -3 + 1 \times 1}{-1}, \frac{2 \times 5 + 1 \times -2}{-1} \right)$

$P_4 = (9, -16)$ (NB: 1 for correct internal division)

(d) $2x-y=0$ $x-2y=0$

$y=2x$ $2y=x$

$m_1=2$ $m_2=\frac{1}{2}$

angle: $\tan \theta = \left| \frac{2 - \frac{1}{2}}{1 + 2 \times \frac{1}{2}} \right|$

$= \left| \frac{\frac{3}{2}}{2} \right|$

obtain: $\tan \theta = \frac{3}{4}$

$\therefore \theta = 143.7^\circ$

$\theta = 143^\circ$ nearest degree

(e) $(n+3)\left(\frac{1}{n} - 2n\right)$

$= 1 - 2n + \frac{3}{n} - 6n^2$

$= 1 - 2n + 3n^{-1} - 6n^2$

①

$\frac{1}{2n} (1 - 2n + 3n^{-1} - 6n^2)$
 $= -2 - \frac{3}{2}n^{-1} - 3n^2$
 $= -2 - \frac{3}{2\sqrt{n}} - \frac{3}{\sqrt{n}}$

①

2(a) $\sin 2A = \frac{1}{2}$

$2 \sin A \cos A = \frac{1}{2}$

$\sin A \cos A = \frac{1}{4}$

$\frac{1}{\sin A \cos A} = \frac{1}{\frac{1}{4}}$

$= 4$

①

b) (i) $\alpha + \beta + \gamma = -\frac{b}{a} = \frac{6}{1} = 6$

$\alpha + \beta + \gamma = 4$

① all 3 correct

$\alpha\beta\gamma = -2$

(ii) $(\alpha-2)(\beta-2)(\gamma-2)$

$= \alpha\beta\gamma - 2(\alpha\beta + \alpha\gamma + \beta\gamma) + 4(\alpha + \beta + \gamma) - 8$

$= -2 - 8 + 24 - 8$

$= 6$

①

c) $\sum_{r=1}^n \frac{1}{r(r+1)} = \frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \frac{1}{3 \cdot 4} + \dots + \frac{1}{n(n+1)}$

Testing $n=1$
 $LHS = \frac{1}{1 \cdot 2}$ $RHS = \frac{1}{1+1}$

$= \frac{1}{2}$ $= \frac{1}{2}$

①

$\therefore$ True for $n=1$
 Assume true for $n=k$

$\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \dots + \frac{1}{k(k+1)} = \frac{k}{k+1}$

①

For $n=k+1$
 $\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \dots + \frac{1}{k(k+1)} + \frac{1}{(k+1)(k+2)} = \frac{k}{k+1} + \frac{1}{(k+1)(k+2)}$

$\sum_{r=1}^k \frac{1}{r(r+1)} + \frac{1}{(k+1)(k+2)} = \frac{k(k+2)+1}{(k+1)(k+2)}$

$= \frac{k^2+2k+1}{(k+1)(k+2)}$

$= \frac{(k+1)^2}{(k+1)(k+2)}$

$= \frac{k+1}{k+2}$

$= \frac{k+1}{k+2}$

$= \frac{k+1}{k+2}$

$= \frac{k+1}{k+2}$

$= \frac{k+1}{k+2}$

①

which is of the form $\frac{n}{n+1}$ where $n=k+1$

$\therefore$ If true for $n=k$, proved true for $n=k+1$
 but is true for $n=1 \therefore$ true for $n=2$
 Is true for $n=2$, is true for $n=3$ and so on
 for all $n \geq 1$

$\therefore \sum_{r=1}^n \frac{1}{r(r+1)} = \frac{n}{n+1}$

① conclusion

(d) $= \frac{x}{y} - \frac{y}{x}$

$\frac{x-y}{xy}$

$= \frac{x^2-y^2}{xy(x-y)}$ (multiplying numerator and denominator by xy)

$= \frac{(x+y)(x-y)}{xy(x-y)}$

$= \frac{x+y}{xy}$

$= \frac{x+y}{xy}$

①

3(a) NB - 1 if diagram not copied from either (ii) (iii) or (iv)

(ii) Angle between a tangent and a chord drawn to the point of contact is equal to the angle in the alternate segment.

①

(iii) Angles at the circumference standing on the same arc are equal

①

(iv) $\angle MAD = \angle AOB$ (alternate angles $MA \parallel BO$)

$\therefore \angle ACD = \angle AOB$ from (i)

①

and since $\angle ACB = \angle AOB$ from (ii)

$\therefore \angle ACD = \angle ACB$

$\therefore AC$ bisects $\angle BCD$

①

b(i) $LHS = 1 - \frac{2}{x-2}$

$= \frac{x-2-2}{x-2}$

$= \frac{x-4}{x-2}$

$= \frac{x-4}{x-2}$

$= \frac{x-4}{x-2}$

$= RHS$

$\therefore 1 - \frac{2}{x-2} = \frac{x-4}{x-2}$

$= \frac{x-4}{x-2}$

①

(ii) $f'(x) = \frac{(x-2) \cdot 1 - (x-4) \cdot 1}{(x-2)^2}$

$= \frac{x-2-x+4}{(x-2)^2}$

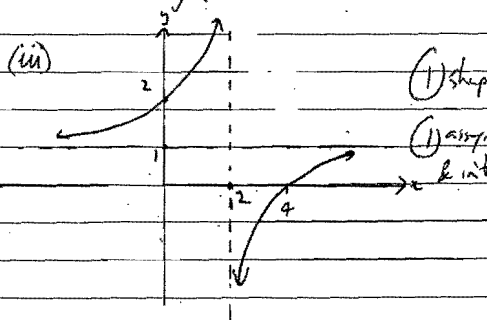
$= \frac{2}{(x-2)^2}$

$f'(x) = \frac{2}{(x-2)^2}$

①

$f'(x) > 0$ since $(x-2)^2 > 0$
 $\therefore$ Increasing for all x in its domain

①



① shape

① asymptote

① x-intercept

when $x=0$, $y=2$

$y=0$, $1 = \frac{2}{x-2}$

$x-2=2$

$x=4$

$x=4$

$$c) f'(x) = \lim_{h \rightarrow 0} \frac{1}{h} \left(\frac{1}{x+h} - \frac{1}{x} \right)$$

$$= \lim_{h \rightarrow 0} \frac{x - (x+h)}{h(x+h)}$$

$$= \lim_{h \rightarrow 0} \frac{-1}{x(x+h)}$$

$$= \lim_{h \rightarrow 0} \frac{-1}{x^2}$$

$$f'(x) = \frac{-1}{x^2}$$

4 a(i) $\sqrt{3} \sin x + \cos x \rightarrow R = \sqrt{3+1} = 2$

$$2 \left(\frac{\sqrt{3}}{2} \sin x + \frac{1}{2} \cos x \right)$$

$$= 2 (\sin x \cos \alpha + \cos x \sin \alpha)$$

$$\therefore \cos \alpha = \frac{\sqrt{3}}{2} \text{ \& } \sin \alpha = \frac{1}{2}$$

$$\alpha = 30^\circ$$

$$\therefore \sqrt{3} \sin x + \cos x = 2 \sin(x+30^\circ)$$

(ii) Max value of $2 \sin(x+30^\circ) = 2 \times 1 = 2$

$$\sqrt{3} \sin x + \cos x = 0 \text{ for } A, 2 \sin(x+30^\circ) = 2$$

$$2 \sin(x+30^\circ) = 0 \quad \sin(x+30^\circ) = 1$$

$$\sin(x+30^\circ) = 0 \quad x+30^\circ = 90^\circ$$

$$x+30^\circ = 0, 180, 360^\circ \quad x = 60^\circ$$

$$x = -30^\circ, 150^\circ, 330^\circ \therefore A \text{ is } 60^\circ$$

First intercept is 150°

$$\therefore B \text{ is } (150^\circ, 0)$$

b(ii) $\tan(45^\circ + A) = \frac{\sin(45^\circ + A)}{\cos(45^\circ + A)}$

$$= \frac{\sin 45^\circ \cos A + \cos 45^\circ \sin A}{\cos 45^\circ \cos A - \sin 45^\circ \sin A}$$

$$= \frac{\frac{\cos A}{\sqrt{2}} + \frac{\sin A}{\sqrt{2}}}{\frac{\cos A}{\sqrt{2}} - \frac{\sin A}{\sqrt{2}}}$$

$$= \frac{\cos A + \sin A}{\cos A - \sin A}$$

$$= \frac{\cos A + \sin A}{\cos A - \sin A}$$

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$$= \frac{\cos A + \sin A}{\cos A - \sin A}$$

$$= \frac{\cos A + \sin A}{\cos A - \sin A}$$

$$= \frac{\cos^2 A + \sin^2 A + 2 \sin A \cos A}{\cos^2 A - \sin^2 A}$$

$$\tan(45^\circ + A) = \frac{1 + \sin 2A}{\cos 2A}$$

(c) $\cos^2 x + \sin^2 2x = 0$

$$\cos^2 x + 2 \sin x \cos x = 0$$

$$\cos x (\cos x + 2 \sin x) = 0$$

$$\cos x = 0 \quad 2 \sin x + \cos x = 0$$

$$2 \sin x = -\cos x$$

$$\cos x = 0 \quad \tan x = -\frac{1}{2}$$

$$x = 90^\circ, 270^\circ \quad x = 180^\circ - 27^\circ, 360^\circ - 27^\circ$$

$$x = 153^\circ, 333^\circ$$

$$\therefore x = 90^\circ, 153^\circ, 270^\circ, 333^\circ$$

5(a) In $\triangle PAB$ and $\triangle PCB$

$$\angle BPA = \angle BPC \text{ (common)}$$

$$\angle PBC = \angle SPB \text{ (alt angles, } ST \parallel BC)$$

$$\therefore \angle PBC = \angle PAB \text{ (since } \angle SPB = \angle PAB \text{ given)}$$

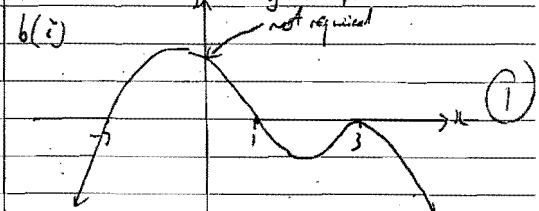
$$\therefore \angle PBA = \angle PCB \text{ (angle sum of } \triangle PAB, \triangle PCB)$$

$$\therefore \triangle PBA \parallel \triangle PCB \text{ (three pairs of matching angles are equal)}$$

(ii) $\frac{PA}{PB} = \frac{PB}{PC}$ (matching sides in similar)

$$\frac{PB}{PC} = \frac{PA}{PC}$$

$$PB^2 = PA \cdot PC$$



$$\text{As } x \rightarrow \infty, P(x) \rightarrow -\infty$$

$$\text{When } x=0, P(0) = 7 \times 10 = 63$$

(iii) From graph

$$7 < x < 1$$

5(c) $\alpha + \beta + \gamma = \frac{\pi}{3}$

$$2\alpha + 2\beta + 2\gamma = \frac{2\pi}{3}$$

$$2\alpha + 2\beta + 2\gamma = \frac{2\pi}{3}$$

$$2\alpha + 2\beta + 2\gamma = 2(\alpha + \beta + \gamma)$$

$$= \frac{2\pi}{3}$$

$$\frac{2}{3} \alpha + \frac{2}{3} \beta + \frac{2}{3} \gamma = \frac{2}{3} (\alpha + \beta + \gamma)$$

$$= \frac{2}{3} \times \frac{\pi}{3}$$

$$2\alpha + 2\beta + 2\gamma = 8\alpha + 8\beta + 8\gamma$$

$$= -8$$

$$\therefore x^2 - 8x + \frac{8}{3}x + \frac{8}{3} = 0$$

$$\therefore 3x^2 - 8x^2 + 8x + 8 = 0$$

d(i) GP with $a=a, r=-\frac{a}{p} = -p$

$$S_{2k+1} = \frac{a(1 - (-p)^{2k+1})}{1 - (-p)}$$

$$= \frac{a(1 - (-1)^{2k+1} p^{2k+1})}{1 + p}$$

$$= \frac{a(1 + p^{2k+1})}{1 + p}$$

(ii) $3^1 - 3^{n+1} + \dots + 1(3)^n = 3^1 - 3^{n+1} + \dots + 3^1 \cdot 3^n$

Here $a=3^n, p=3$ and $n=k$

$$\therefore S_{2k+1} = \frac{3^n(1 + 3^{2k+1})}{1 + 3}$$

$$= \frac{3^n(1 + 3^{2k+1})}{4}$$

6(a) $x^2 + y^2 + 2xy - z^2$

$$= (x+y)^2 - z^2$$

$$= (x+y+z)(x+y-z)$$

b) Let $\angle OAG = y$

$$\angle OAG = \angle OAG \text{ (given AF bisects } \angle OAC)$$

$$\angle DOA = 2y \text{ (base angles of isosceles } \triangle OAD)$$

$$\angle OBA = y \text{ (angle at circumference is half angle at centre standing on same arc)}$$

$$\therefore \angle OBA = \angle OAG = y$$

$$\therefore \triangle GAB \text{ is isosceles (base angles equal)}$$

$$\therefore GA = GB \text{ (equal sides opposite equal angles in isosceles } \triangle GAB)$$

(ii) Join AE

$$\angle AGE = 2y \text{ (exterior } \angle \text{ of } \triangle OAG)$$

$$\therefore \angle DOA = \angle AGE = 2y$$

$$\therefore AOC \text{ is a cyclic quadrilateral (angles at the circumference standing on the same arc are equal)}$$

(iii) Join CE

$$\angle ACE = \angle ABE \text{ (angles at circumference standing on the same arc)}$$

$$\angle ACE = \angle FAC \text{ (} \angle FAC = y \text{ proved above)}$$

$$\therefore CE \parallel AF \text{ (alternate angles are equal)}$$

(c) Test $n=1$

$$5 + 2 \times 1 = 27$$

$$= 3 \times 9$$

$$\therefore \text{Divisible by 3}$$

$$\therefore \text{True for } n=1$$

Assume true for $n=k$

$$5^k + 2(11^k) = 3P \text{ where } P \text{ is an integer} \quad (1)$$

for $n=k+1$

$$5^{k+1} + 2(11^{k+1}) = 5 \cdot 5^k + 2 \cdot 11 \cdot 11^k$$

$$= 5(5^k - 2(11^k)) + 22 \cdot 11^k \quad (1)$$

$$= 15P - 10 \cdot 11^k + 22 \cdot 11^k$$

$$= 15P + 12 \cdot 11^k$$

$$= 3(5P + 4 \cdot 11^k)$$

$$= 3 \times Q \text{ where } Q \text{ is an integer} \quad (1)$$

(since $5P$ and 11^k are integers)

which is divisible by 3.

$\therefore$ If true for $n=k$, proved true for $n=k+1$

but true for $n=1$, $\therefore$ true for $n=2$

but true for $n=2$, $\therefore$ true for $n=3$ and so on for all $n \geq 1$

$\therefore 5^n + 2(11^n)$ is divisible by 3

$$7.1(i) \tan A = \frac{h}{AC}$$

$$\tan 20^\circ = \frac{h}{AC}$$

$$AC = \frac{h}{\tan 20^\circ}$$

$$\text{Similarly } BC = \frac{h}{\tan 10^\circ}$$

for both

(ii) In $\triangle ABC$

$$BC^2 = AB^2 + AC^2$$

$$AB^2 = BC^2 - AC^2$$

$$40^2 = h^2 \cot^2 10^\circ - h^2 \cot^2 20^\circ \quad (1)$$

$$1600 = h^2 (\cot^2 10^\circ - \cot^2 20^\circ)$$

$$h^2 = \frac{1600}{\cot^2 10^\circ - \cot^2 20^\circ}$$

$$\therefore h = \frac{40}{\sqrt{\cot^2 10^\circ - \cot^2 20^\circ}} \quad (1)$$

$$(iii) h = \frac{40}{\sqrt{24.16}}$$

$$\therefore \text{Height} = 8.06 \text{ m} \quad (1)$$

$$7(b)(i) A_1 = PR - m$$

$$A_2 = (PR - m)R - m$$

$$A_2 = PR^2 - mR - m$$

(Vetter)

$$(ii) A_3 = A_2 R - m$$

$$= PR^2 - mR^2 - mR - m$$

$$= PR^3 - m(1 + R + R^2)$$

A pattern is emerging

$$A_n = PR^n - m(1 + R + R^2 + \dots + R^{n-1}) \quad (1)$$

$$= PR^n - m \left(\frac{1(R^n - 1)}{R - 1} \right)$$

$$A_n = PR^n - \frac{m(R^n - 1)}{R - 1} \quad (1)$$

$$(iii) \text{ If } A_n = \frac{K}{100} \times P$$

$$\frac{PK}{100} = PR^n - \frac{m(R^n - 1)}{R - 1} \quad (1)$$

$$PK(R - 1) = 100(R - 1)PR^n - 100m(R^n - 1)$$

$$R^n(100P(R - 1) - 100m) = PK(R - 1) - 100m \quad (1)$$

$$R^n(100(P(R - 1) - m)) = PK(R - 1) - 100m$$

$$\therefore R^n = \frac{PK(R - 1) - 100m}{100[P(R - 1) - m]} \quad (1)$$

(iv) From calculator,

$$R^n = \frac{40000 \times 20 \left(\frac{1}{500} \right) - 100 \times 800}{100 \left[40000 \times \frac{1}{500} - 800 \right]}$$

$$R^n = 1.48 \quad (1)$$

$$1.0075^n = 1.48 \quad (\text{then } (1) \text{ for } 4 \text{ or } 5)$$

when $n = 48$, $1.0075^{48} = 1.43$ when $n = 60$, $1.0075^{60} = 1.565$
 $\therefore$ between 4 and 5 yrs.